TECHNICAL SPECIFICATION

GuestFlow.ge

სტუმართა მართვის პლატფორმა

All-in-one booking, guest management & channel sync
for Georgian guesthouses and small hotels

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1. Executive Summary

GuestFlow.ge is a Georgian-language SaaS platform purpose-built for small guesthouses, family hotels, and boutique accommodations across Georgia. The platform provides an affordable, localized alternative to enterprise property management systems like Cloudbeds or Mews, which are too expensive, English-only, and feature-bloated for a typical 3–15 room Georgian guesthouse.

The platform combines room and availability management, direct booking engine, channel manager (Booking.com and Airbnb synchronization), guest communication with AI-powered translation, financial reporting in GEL, and local payment integration (BOG iPay, TBC Pay) into a single, intuitive dashboard fully localized in Georgian.

1.1 Market Opportunity

- Georgia's tourism sector accounts for ~15% of GDP and is growing steadily
- Thousands of small guesthouses operate across Kakheti, Svaneti, Gudauri, Mestia, Batumi, and rural regions
- Only 7% of Georgian SMEs have sophisticated websites (vs 78% EU average)
- Most guesthouses manage bookings via WhatsApp, phone calls, and paper notebooks
- No Georgian-language property management tool exists at an affordable price point
- Government (GITA) offers 10-year tax breaks for innovative startups

1.2 Key Differentiators

- Full Georgian language interface (ქართული ინტერფეისი)
- GEL-native: pricing, invoices, reports, and tax calculations in Georgian Lari
- Local payment integration: BOG iPay, TBC Pay, Liberty Pay
- AI-powered message translation: Georgian ↔ English/Russian/German/Turkish
- Pricing designed for Georgian market: starting at ₾49/mo (vs $100+/mo for global tools)
- Mobile-first design: works on any smartphone (many rural guesthouse owners rely on mobile)

2. System Architecture

2.1 High-Level Components

Component	Technology	Responsibility
Frontend SPA	React 18 + TypeScript + Vite	Dashboard UI, booking calendar, channel management
API Server	Django 5.x + Django REST Framework	Business logic, auth, multi-tenancy, API endpoints
Channel Manager	Python + Celery workers	Booking.com & Airbnb API sync, rate management
AI Translation Service	Python + OpenAI API	Guest message translation (GE/EN/RU/DE/TR)
Notification Service	Celery + Twilio + SendGrid	SMS, WhatsApp, email notifications to guests & owners

Task Queue	Celery + Redis	Async channel sync, notifications, report generation
Database	PostgreSQL 16 (Supabase)	Multi-tenant data, bookings, guests, properties
Object Storage	Supabase Storage	Property photos, guest documents, invoices
Cache	Redis (Upstash)	Session cache, rate limiting, availability cache
Payment Gateway	BOG iPay / TBC Pay APIs	GEL payment processing for direct bookings

2.2 Multi-Tenancy Model

Each guesthouse operates as an isolated tenant. Data isolation is enforced at the database level using a tenant_id foreign key on all models. The tenant is resolved from the JWT token on every API request. This approach (shared database, shared schema, tenant column) is the simplest for a solo developer and scales well to thousands of tenants.

2.3 Communication Patterns

- Frontend ↔ Backend: REST API over HTTPS with JWT (access + refresh tokens)
- Channel sync: Celery periodic tasks polling Booking.com/Airbnb APIs every 5 minutes
- Real-time updates: Server-Sent Events (SSE) for new booking notifications on the dashboard
- Guest notifications: SMS via Twilio, email via SendGrid, WhatsApp via Twilio Business API
- AI translation: async Celery tasks calling OpenAI API, results cached per message

3. Frontend Architecture

3.1 Tech Stack

Library	Version	Purpose
React	18.x	UI framework
TypeScript	5.x	Type safety
Vite	5.x	Build tool, dev server, HMR
React Router	6.x	Client-side routing
TanStack Query	5.x	Server state, caching, optimistic updates
Zustand	4.x	Client state (UI state, locale, sidebar)
Tailwind CSS	3.x	Utility-first styling
shadcn/ui	—	Component library (Radix primitives)
react-big-calendar	1.x	Visual booking calendar
react-i18next	14.x	Georgian/English internationalization
React Hook Form + Zod	—	Form handling and validation

date-fns	3.x	Date manipulation (timezone-aware)

3.2 Internationalization (i18n)

The entire UI is localized in Georgian as the primary language, with English as a secondary option. All labels, error messages, date formats, and currency displays use Georgian conventions. Date format: DD.MM.YYYY. Currency: ₾ (GEL) with Georgian number formatting. The react-i18next library manages translation keys stored in /locales/ka.json and /locales/en.json.

3.3 Application Routes

Route	Page	Auth
/	Marketing landing page (Georgian)	No
/login	Authentication (email + password)	No
/register	Property registration wizard	No
/book/:slug	Public booking page for guests	No
/dashboard	Main dashboard (today's arrivals, occupancy, revenue)	Yes
/calendar	Visual booking calendar (drag-and-drop)	Yes
/bookings	Booking list with filters and search	Yes
/bookings/:id	Booking detail (guest info, payment, notes)	Yes
/rooms	Room management (types, pricing, photos)	Yes
/guests	Guest database with search	Yes
/channels	Channel manager (Booking.com, Airbnb connection)	Yes
/messages	Guest messaging with AI translation	Yes
/reports	Financial reports (revenue, occupancy, tax)	Yes
/settings	Property settings, billing, team access	Yes
/settings/billing	Subscription management (Stripe/local)	Yes

3.4 Key UI Screens

3.4.1 Dashboard

The main dashboard shows at-a-glance metrics: today's check-ins and check-outs, current occupancy percentage, this month's revenue in GEL, unread guest messages, and pending channel sync status. A timeline view shows upcoming arrivals for the next 7 days. All data refreshes via TanStack Query polling every 30 seconds.

3.4.2 Booking Calendar

A visual calendar grid showing rooms on the Y-axis and dates on the X-axis. Bookings appear as colored bars spanning check-in to check-out dates. Color coding: blue for confirmed, yellow for pending, green for checked-in, gray for checked-out, red for cancelled. Drag-and-drop to move bookings between rooms. Click-to-create for new manual bookings. Source badges (Booking.com, Airbnb, Direct) appear on each booking bar.

3.4.3 Guest Messaging

A WhatsApp-style chat interface showing conversations with guests. Each message has a translate button that triggers AI translation. The owner types in Georgian, the system translates and sends in the guest's language (and vice versa). Supports pre-built quick-reply templates: check-in instructions, directions, WiFi password, house rules — all in multiple languages.

3.4.4 Public Booking Page

Each property gets a public booking page at guestflow.ge/book/:slug. Shows property photos, room types with prices in GEL, availability calendar, and a booking form. Supports Georgian, English, and Russian via language toggle. Integrated with BOG iPay/TBC Pay for direct GEL payments. Mobile-optimized for tourists browsing on phones.

4. Backend Architecture

4.1 Django Apps

App	Responsibility
accounts	User registration, JWT auth, property owner profiles, team invitations
properties	Property CRUD, room types, amenities, photos, pricing rules, seasonal rates
bookings	Booking lifecycle (create, confirm, check-in, check-out, cancel), availability engine
guests	Guest profiles, contact info, visit history, preferences, document storage
channels	Booking.com & Airbnb API integration, availability sync, rate push, reservation import
messaging	Guest conversations, AI translation, quick-reply templates, notification dispatch
payments	BOG iPay / TBC Pay integration, payment tracking, refunds, invoice generation
reports	Revenue reports, occupancy analytics, tax summaries, export to Excel/PDF
billing	SaaS subscription management, plan enforcement, usage tracking
notifications	SMS, email, WhatsApp dispatch via Twilio/SendGrid, notification preferences

4.2 Database Schema

4.2.1 Property

Field	Type	Description
id	UUID	Primary key
owner_id	FK → User	Property owner
name_ka	VARCHAR(255)	Property name in Georgian
name_en	VARCHAR(255)	Property name in English
slug	VARCHAR(100)	URL slug for public booking page
property_type	ENUM(guesthouse, hotel, apartment, villa)	Accommodation type
address_ka	TEXT	Address in Georgian
city	ENUM(tbilisi, batumi, kutaisi, gudauri, ...)	City/region for search
latitude	DECIMAL(10,8)	GPS latitude
longitude	DECIMAL(11,8)	GPS longitude
phone	VARCHAR(20)	Contact phone (Georgian mobile format)
whatsapp	VARCHAR(20)	WhatsApp number for guest communication
check_in_time	TIME	Default check-in time
check_out_time	TIME	Default check-out time
house_rules_ka	TEXT	House rules in Georgian
house_rules_en	TEXT	House rules in English
tax_id	VARCHAR(20)	Georgian tax identification number
plan	ENUM(starter, pro, business)	Subscription tier
is_active	BOOLEAN	Whether property is live
created_at	TIMESTAMP	Registration date

4.2.2 Room

Field	Type	Description
id	UUID	Primary key
property_id	FK → Property	Parent property
name_ka	VARCHAR(100)	Room name in Georgian (e.g., სოთახი 201)
name_en	VARCHAR(100)	Room name in English
room_type	ENUM(single, double, twin, family, suite, dorm)	Room category
max_guests	INTEGER	Maximum occupancy

base_price_gel	DECIMAL(10,2)	Base nightly rate in GEL
amenities	JSONB	Array of amenity codes (wifi, parking, ac, balcony, etc.)
photos	JSONB	Array of photo storage keys
floor	INTEGER	Floor number
is_active	BOOLEAN	Available for booking
sort_order	INTEGER	Display order

4.2.3 Booking

Field	Type	Description
id	UUID	Primary key
property_id	FK → Property	Property
room_id	FK → Room	Assigned room
guest_id	FK → Guest	Guest profile
source	ENUM(direct, booking_com, airbnb, phone, walkin)	Booking channel
external_id	VARCHAR(100)	Booking.com/Airbnb reservation ID
status	ENUM(pending, confirmed, checked_in, checked_out, cancelled, no_show)	Booking status
check_in	DATE	Check-in date
check_out	DATE	Check-out date
num_guests	INTEGER	Number of guests
total_price_gel	DECIMAL(10,2)	Total price in GEL
paid_amount_gel	DECIMAL(10,2)	Amount already paid
payment_status	ENUM(unpaid, partial, paid, refunded)	Payment tracking
payment_method	ENUM(cash, bog_ipay, tbc_pay, bank_transfer, channel)	How payment was made
notes_ka	TEXT	Internal notes in Georgian
guest_message	TEXT	Message from guest at booking time
created_at	TIMESTAMP	When booking was created

4.2.4 Guest

Field	Type	Description
id	UUID	Primary key

property_id	FK → Property	Tenant scope
first_name	VARCHAR(100)	Guest first name
last_name	VARCHAR(100)	Guest last name
email	VARCHAR(255)	Email address
phone	VARCHAR(20)	Phone number
country	VARCHAR(2)	ISO country code
language	VARCHAR(2)	Preferred language (ka, en, ru, de, tr)
id_number	VARCHAR(50)	Passport or ID number
total_stays	INTEGER	Number of visits (denormalized)
total_spent_gel	DECIMAL(10,2)	Lifetime spend in GEL
notes	TEXT	Internal notes about the guest
created_at	TIMESTAMP	First visit date

4.2.5 SeasonalRate

Field	Type	Description
id	UUID	Primary key
room_id	FK → Room	Room this rate applies to
name_ka	VARCHAR(100)	Season name (e.g., ზაფხული სეზონი)
start_date	DATE	Season start
end_date	DATE	Season end
price_gel	DECIMAL(10,2)	Override nightly rate in GEL
min_stay	INTEGER	Minimum night stay during this season

4.2.6 ChannelConnection

Field	Type	Description
id	UUID	Primary key
property_id	FK → Property	Property
channel	ENUM(booking_com, airbnb)	OTA platform
api_key	TEXT	Encrypted API credentials
property_external_id	VARCHAR(100)	Property ID on the channel
sync_status	ENUM(active, paused, error)	Current sync state
last_sync_at	TIMESTAMP	Last successful sync
room_mappings	JSONB	Map of local room_id → channel room_id

5. API Endpoints

5.1 Authentication

Method	Endpoint	Description
POST	/api/auth/register/	Register property owner + create property
POST	/api/auth/login/	JWT token pair
POST	/api/auth/refresh/	Refresh access token
GET	/api/auth/me/	Current user profile

5.2 Properties

Method	Endpoint	Description
GET	/api/properties/me/	Current user's property details
PATCH	/api/properties/me/	Update property settings
POST	/api/properties/me/photos/	Upload property photos

5.3 Rooms

Method	Endpoint	Description
GET	/api/rooms/	List all rooms
POST	/api/rooms/	Create room
PATCH	/api/rooms/:id/	Update room details/pricing
DELETE	/api/rooms/:id/	Deactivate room
GET	/api/rooms/:id/availability/	Room availability for date range
POST	/api/rooms/:id/rates/	Create seasonal rate

5.4 Bookings

Method	Endpoint	Description
GET	/api/bookings/	List bookings (filterable by status, date, source)
POST	/api/bookings/	Create manual booking
GET	/api/bookings/:id/	Booking detail
PATCH	/api/bookings/:id/	Update booking
POST	/api/bookings/:id/check-in/	Mark guest as checked in
POST	/api/bookings/:id/check-out/	Mark guest as checked out
POST	/api/bookings/:id/cancel/	Cancel booking
POST	/api/bookings/:id/payment/	Record payment

GET	/api/bookings/calendar/	Calendar view data (date range query)

5.5 Guests

Method	Endpoint	Description
GET	/api/guests/	List guests (searchable)
GET	/api/guests/:id/	Guest profile with booking history
PATCH	/api/guests/:id/	Update guest info

5.6 Channel Manager

Method	Endpoint	Description
GET	/api/channels/	List connected channels
POST	/api/channels/connect/	Initiate channel connection (OAuth flow)
DELETE	/api/channels/:id/	Disconnect channel
POST	/api/channels/:id/sync/	Force manual sync
GET	/api/channels/:id/status/	Sync status and last errors
PATCH	/api/channels/:id/mappings/	Update room mappings

5.7 Messaging

Method	Endpoint	Description
GET	/api/messages/	List conversations
GET	/api/messages/:guest_id/	Conversation with specific guest
POST	/api/messages/:guest_id/	Send message
POST	/api/messages/:id/translate/	Translate a message (AI)
GET	/api/messages/templates/	List quick-reply templates

5.8 Reports

Method	Endpoint	Description
GET	/api/reports/revenue/	Revenue report (daily/weekly/monthly)
GET	/api/reports/occupancy/	Occupancy rate report
GET	/api/reports/sources/	Bookings by source breakdown
GET	/api/reports/tax-summary/	Tax summary for Georgian tax filing
GET	/api/reports/export/	Export report as Excel or PDF

5.9 Public Booking API

Method	Endpoint	Description
GET	/api/public/:slug/	Property info for public booking page
GET	/api/public/:slug/availability/	Available rooms for date range
POST	/api/public/:slug/book/	Create direct booking (guest-facing)
POST	/api/public/:slug/pay/	Initiate BOG iPay / TBC Pay payment
GET	/api/public/:slug/pay/callback/	Payment callback/confirmation

6. Channel Manager Integration

6.1 Booking.com Connectivity Partner API

GuestFlow connects to Booking.com via their Connectivity Partner API. This requires applying to the Booking.com Connectivity Partner Program (free, but requires review). The integration handles: pulling new reservations, pushing availability and rate updates, sending booking confirmations and modifications, and syncing room mappings.

Sync frequency: every 5 minutes via Celery Beat periodic tasks. On each sync cycle, the worker checks for new/modified reservations, pushes local availability changes, and reconciles rate overrides.

6.2 Airbnb API (Host Tools Partner)

Airbnb integration uses the Host Tools API. Similar to Booking.com, this requires partner application. The integration syncs: listings (room → listing mapping), availability calendar, pricing, and reservation data. Airbnb uses a pull model for new reservations combined with webhooks for real-time notification of changes.

6.3 Availability Engine

The availability engine is the critical component preventing double-bookings across channels. When a booking is created from any source (direct, Booking.com, Airbnb, manual), the engine immediately: marks the room as unavailable for those dates in the local database, queues availability push tasks to all connected channels, and verifies no overlap exists (database-level constraint: unique index on room_id + date for each occupied night). The engine uses a date-range occupancy table (RoomNight model) with a unique constraint on (room_id, date) to make double-bookings physically impossible at the database level.

6.4 Rate Management

Rates flow from GuestFlow to channels, not the other way. The owner sets base rates and seasonal rates in GuestFlow, and these are pushed to Booking.com and Airbnb. Per-channel rate adjustments are supported (e.g., Booking.com rate = base + 15% commission markup). This ensures GuestFlow is the single source of truth for pricing.

7. AI-Powered Features

7.1 Guest Message Translation

The core AI feature: real-time translation of guest messages between Georgian and English, Russian, German, and Turkish. When a guesthouse owner receives a message in English, they see the original plus a Georgian translation. When they reply in Georgian, the system translates to the guest's language before sending. Powered by OpenAI GPT-4o with hospitality-specific system prompts for accurate context-aware translation.

- Translation is async via Celery to avoid blocking the UI
- Results are cached per message (translate once, display forever)
- System prompt includes hospitality terminology and Georgian cultural context
- Cost: ~$0.001 per message translation (negligible at scale)

7.2 AI Listing Description Generator

When a property owner adds photos and basic info about their guesthouse, AI generates professional listing descriptions in Georgian and English. Input: property type, location, amenities, room details, photos. Output: compelling marketing copy optimized for Booking.com and Airbnb listing formats.

7.3 Dynamic Pricing Suggestions

AI analyzes occupancy patterns, seasonal trends, and local events to suggest optimal nightly rates. The system learns from the property's booking history and compares against area averages. Suggestions are displayed as a dashboard widget — the owner always has final control over pricing.

7.4 Smart Review Responses

When a guest leaves a review on Booking.com or Airbnb, AI drafts a professional response in the appropriate language. Positive reviews get a warm thank-you. Negative reviews get an empathetic, solution-oriented response. The owner reviews and edits before posting.

8. Payment Integration

8.1 Local Payment Gateways

Gateway	Use Case	Integration Method
BOG iPay	Direct bookings via public page	REST API with redirect-based checkout
TBC Pay	Alternative payment option	REST API with redirect-based checkout
Bank Transfer	Manual payment for walk-ins	Owner records manually in system
Cash	On-arrival payment	Owner records at check-in

8.2 Payment Flow (Direct Booking)

1. Guest selects room and dates on public booking page
2. Guest fills contact info and selects payment method (BOG iPay or TBC Pay)
3. System creates booking with status=pending and redirects to payment gateway
4. Guest completes payment on the bank's page
5. Gateway sends callback to /api/public/:slug/pay/callback/
6. System verifies payment, updates booking to status=confirmed
7. Confirmation SMS/email sent to guest and WhatsApp notification to owner

8.3 SaaS Subscription Billing

GuestFlow's own subscription billing uses Stripe for card payments (international standard) with a local bank transfer option for Georgian businesses that prefer it. The billing cycle is monthly with annual discount option.

9. Pricing Model

Plan	Price	Rooms	Key Features
Starter	₾49/mo	1–5	Booking calendar, manual bookings, guest management, GEL invoices, public booking page, SMS notifications
Pro	₾99/mo	6–20	All Starter + Booking.com sync, Airbnb sync, AI translation, seasonal rates, financial reports, WhatsApp notifications
Business	₾199/mo	21–50	All Pro + dynamic pricing AI, multi-user access (3 seats), priority support, custom booking page domain, API access

Revenue targets: 100 properties at average ₾80/mo = ₾8,000/mo (~$2,900 USD). 250 properties = ₾20,000/mo (~$7,200 USD). The Georgian guesthouse market has 2,000–5,000 potential customers.

10. Infrastructure & Deployment

10.1 Hosting

Component	Platform	Rationale
React Frontend	Vercel	Global CDN, preview deploys, zero-config
Django API	Railway	Easy container deploy, managed scaling, EU region (low latency to GE)
Celery Workers	Railway	Same platform as API, auto-scaling workers
PostgreSQL	Supabase	Managed Postgres, built-in auth helpers, row-level security
Redis	Upstash	Serverless Redis, low cost, global edge

Object Storage	Supabase Storage	Property photos, generated invoices
Domain	guestflow.ge	.ge domain for local trust and SEO

10.2 CI/CD

- GitHub Actions: frontend (lint + build + deploy to Vercel), backend (lint + test + deploy to Railway)
- Preview environments for pull requests
- Database migrations auto-run on deploy
- Staging environment mirrors production for channel API testing

10.3 Monitoring

- Sentry for error tracking (frontend + backend)
- Uptime monitoring on all API endpoints
- Celery Flower for task queue monitoring
- Channel sync health dashboard (alert if sync fails 3x consecutively)

11. Security

- HTTPS everywhere (TLS 1.3)
- JWT in httpOnly cookies, CSRF protection
- Channel API credentials encrypted at rest (AES-256)
- CORS restricted to guestflow.ge and Vercel preview domains
- Rate limiting: 100 req/min authenticated, 20 req/min anonymous
- Multi-tenant data isolation enforced at query level (every queryset filtered by property_id)
- Guest PII (passport numbers) encrypted at rest
- GDPR-aware: guest data export and deletion endpoints
- Input validation via DRF serializers + Zod on frontend
- SQL injection prevention via Django ORM
- Dependency scanning via Dependabot

12. MVP Roadmap

12.1 Phase 1 — Core Platform (Weeks 1–7)

Week	Milestone	Deliverables
1–2	Project Setup + Auth + DB	Django scaffold, React app, JWT auth, PostgreSQL schema, Property/Room/Booking models, i18n setup (Georgian + English)
3–4	Booking Engine + Calendar	Room CRUD, availability engine, RoomNight model, booking lifecycle, visual calendar (react-big-calendar), manual booking creation, drag-and-drop

5	Guest Management + Messaging	Guest CRUD, search, visit history, messaging UI, quick-reply templates, notification service (SMS + email)
6	Public Booking Page + Payments	Public booking page (/book/:slug), BOG iPay integration, payment flow, booking confirmation, mobile-responsive design
7	Dashboard + Reports + Deploy	Dashboard with occupancy/revenue metrics, basic financial reports, GEL invoice generation, deploy to Vercel + Railway, staging environment

12.2 Phase 2 — Channel Manager + AI (Weeks 8–12)

Week	Milestone	Deliverables
8–9	Booking.com Integration	Connectivity Partner application, API integration, reservation import, availability push, rate sync, room mapping UI
10	Airbnb Integration	Host Tools API, listing sync, reservation import, calendar sync
11	AI Features	Message translation (GPT-4o), listing description generator, review response drafts, translation caching
12	Billing + Polish	Stripe subscription billing, plan enforcement, usage limits, onboarding wizard, error handling, loading states

12.3 Phase 3 — Launch & Growth (Weeks 13–18)

Week	Milestone	Deliverables
13–14	Landing Page + Georgian SEO	Marketing site in Georgian, Google Ads setup (Georgian keywords), Facebook/Instagram presence, guesthouse owner testimonials
15–16	Testing + Hardening	Unit tests (pytest), integration tests, E2E tests (Playwright), security review, channel sync stress testing, 10-property beta test
17–18	Beta + Launch	Invite 20–30 guesthouses from Kakheti/Svaneti, collect feedback, iterate, public launch with ₾29 first-month promo pricing

13. Go-to-Market Strategy

13.1 Launch Strategy

1. Start with Kakheti wine region — highest density of guesthouses, tech-savvier owners, tourism-dependent
2. Partner with local tourism associations and GITA for credibility
3. Offer free 1-month trial + in-person setup assistance (visit 10–20 guesthouses personally)
4. Georgian-language Facebook group for guesthouse owners (build community)
5. YouTube tutorials in Georgian showing how to use the platform
6. Referral program: existing users get 1 free month for each referral

13.2 Sales Channels

- Direct outreach: visit guesthouses in key tourist regions
- Facebook ads targeting guesthouse owners (Georgian language)
- Partnership with Booking.com regional office (they want more Georgian properties online)
- GITA startup programs and events
- Word of mouth within the tight-knit Georgian hospitality community

13.3 Expansion Path

1. Year 1: 100–200 Georgian properties (₾8K–20K/mo revenue)
2. Year 2: Expand to Batumi apartments market + add Airbnb integration polish
3. Year 3: Expand to Armenia and Azerbaijan (similar market conditions, nearby regions)
4. Year 4: White-label option for tourism boards and hotel associations

14. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
Booking.com partner application rejected/delayed	High	Medium	Build MVP with manual booking first; apply to partner program in Phase 1 so it's approved by Phase 2
Low tech adoption by rural guesthouse owners	High	Medium	In-person onboarding, ultra-simple mobile UI, Georgian-language video tutorials, phone support
OpenAI API costs for translation exceed budget	Low	Low	Cache all translations, use cheaper models for simple messages, batch processing
Channel sync causes double-bookings	High	Low	Database-level unique constraint on RoomNight, 5-min sync frequency, manual sync button, instant push on local changes
Payment gateway integration complexity (BOG/TBC)	Medium	Medium	Start with one gateway (BOG iPay), add TBC later; cash/bank transfer as fallback
Solo developer bottleneck	Medium	High	Use AI for code generation, managed services for infrastructure, ruthless feature prioritization
Competitors enter Georgian market	Medium	Low	First-mover advantage, deep local knowledge, Georgian language moat, low pricing

Appendix A: Environment Variables

Variable	Service	Description
DATABASE_URL	Django	PostgreSQL connection string (Supabase)

REDIS_URL	Django/Celery	Redis connection (Upstash)
SECRET_KEY	Django	Django secret key
OPENAI_API_KEY	Django	OpenAI API key for translation and AI features
STRIPE_SECRET_KEY	Django	Stripe key for SaaS billing
STRIPE_WEBHOOK_SECRET	Django	Stripe webhook verification
BOG_IPAY_MERCHANT_ID	Django	Bank of Georgia iPay merchant ID
BOG_IPAY_SECRET	Django	Bank of Georgia iPay secret key
TBC_PAY_CLIENT_ID	Django	TBC Pay client ID
TBC_PAY_SECRET	Django	TBC Pay secret key
TWILIO_ACCOUNT_SID	Django	Twilio account for SMS/WhatsApp
TWILIO_AUTH_TOKEN	Django	Twilio auth token
TWILIO_PHONE_NUMBER	Django	Twilio sender phone number
SENDGRID_API_KEY	Django	SendGrid email delivery key
BOOKING_COM_USERNAME	Channel Mgr	Booking.com API username
BOOKING_COM_PASSWORD	Channel Mgr	Booking.com API password
AIRBNB_CLIENT_ID	Channel Mgr	Airbnb API client ID
AIRBNB_CLIENT_SECRET	Channel Mgr	Airbnb API client secret
SUPABASE_URL	Django	Supabase project URL
SUPABASE_KEY	Django	Supabase service role key
VITE_API_URL	React	Backend API base URL
SENTRY_DSN	All	Sentry error tracking DSN

Appendix B: Georgian-Specific Requirements

B.1 Tax Compliance

- Georgian guesthouses with annual revenue above ₾100,000 must register as VAT payers (18%)
- System generates tax summaries compatible with RS.ge (Revenue Service) reporting format
- Invoice format follows Georgian tax law requirements (tax ID, sequential numbering, GEL amounts)

B.2 Guest Registration

- Georgian law requires accommodation providers to register foreign guests with the Tourism Department
- System collects and stores passport/ID data securely for compliance
- Future feature: automated guest registration submission via government API (when available)

B.3 Cultural Considerations

- Georgian hospitality (სტუმართმოყვარეობა) is deeply personal — the UI must feel warm, not corporate
- Many guesthouse owners are 50+ years old — UI must be extremely simple with large tap targets
- Phone-based workflow: many owners will use the app primarily on their smartphone
- WhatsApp is the primary communication tool in Georgian hospitality — deep WhatsApp integration is essential
- Georgian feast (სუფრა) culture: consider adding meal/dinner booking as a future feature